

# EE 304 Energy Conversion - Spring 2025

## Midterm Project: Sound Production in Space

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### I. INTRODUCTION

This project aims to design a device that allows an astronaut to generate sound in the vacuum of space and transmit it to the International Space Station (ISS). The proposed device, conceptualized as a string instrument, leverages electromagnetic principles to facilitate the conversion of electrical energy into mechanical vibrations, which are subsequently transformed into audible sound. The overall system model for converting sound into electrical signals and reproducing it aboard the ISS is illustrated in Fig. 1.

The process begins when the astronaut plucks the strings of the electromagnetic transducer in space. The resulting vibrations induce an electrical signal through the principle of electromagnetic induction. To optimize signal transmission, high-frequency components beyond the audible range of human perception are attenuated using a RC filter, effectively reducing the signal bandwidth and enhancing overall system efficiency. The weak electrical signals generated by the transducer are then amplified using a preamplifier-amplifier module, ensuring the voltage levels are suitable for wireless transmission. The signal is transmitted via an antenna and propagates through space until it is received by a corresponding receiver antenna on the ISS. Upon reception, the signal undergoes further amplification before being converted back into mechanical vibrations by a loudspeaker, ultimately producing audible sound.

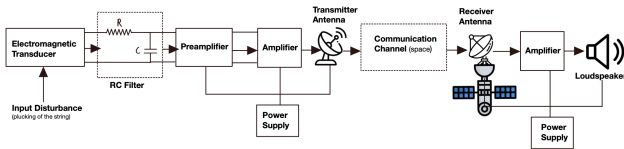


Fig. 1: End-to-end system model producing sound within space station

This study primarily focuses on the electromagnetic design of the input transducer, responsible for converting string vibrations into electrical signals. The following sections present the theoretical foundations, design considerations, and operational guidelines for the proposed device.

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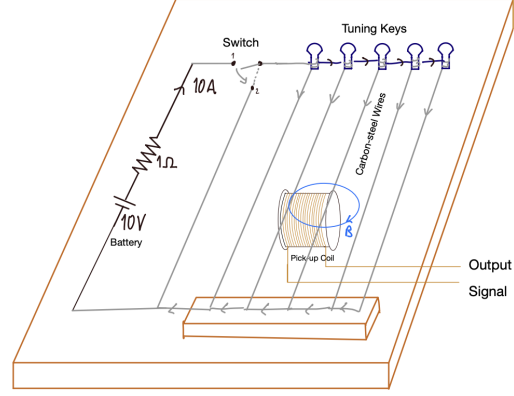


Fig. 2: Schematic of Design Option I comprised of 5 strings of adjustable tension, power circuitry and pick-up coil.

### II. THEORY

In this section, we will explore the theoretical underpinnings of the device, focusing on electromagnetic energy conversion. We will discuss two design options: one involving a magnetized string and another utilizing a permanent magnet.

#### A. Design Option I: Magnetized String

The first design option involves the use of a magnetized string made from the carbon steel (1010) wire. As can be seen from 2 carbon steel wire of 10 m is divided into 5 pieces of  $L = 1.5$  m and each piece stretched through a supporting platform with one end fixed and the other is wrapped around tuning keys. Via tuning keys tension in each cable can be adjusted in order to obtain a variety of fundamental frequencies similar to a guitar. Carbon steel cable is extended to make an auxiliary branch in parallel, with a switch attached at the junction. In another branch parallel to those, a 10 V battery along with a 1  $\Omega$  resistor is placed.

In order to detect fluctuations in magnetic flux a pick-up coil made out of a copper wire is placed between the strings and platform with its symmetry axis parallel to the platform but perpendicular to strings. Pick-up coil is made of a copper wire wound around a nonconducting cylindrical material of  $r = 1$  cm. Available copper wire of 1 m limits the number of windings to  $N_C = 16$  but ideally, we would like to have as many windings as possible in order get a strong signal. Furthermore, it's assumed that coil is wide enough so that flux from each string is independent of its relative position to the pick-up coil.

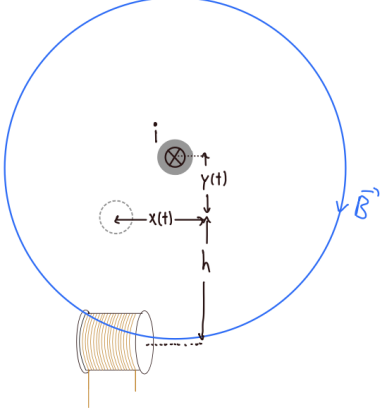


Fig. 3: Magnetic field produced by carbon steel strings and flux created in the pick-up coil.

Current passing through the carbon steel wire will create a circular magnetic field around it. Magnetic field of an effectively infinite wire, since  $r \ll L$ , at a distance  $r$  away from the center of the wire can be easily calculated as

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I_{enc} \quad , \quad \vec{B} = \frac{\mu_0 I}{2\pi r} \hat{\theta} \quad (1)$$

where  $\mu_0$  is the permeability of free space,  $I$  is the constant current passing through each string and  $\hat{\theta}$  is the unit vector in radial direction. Note that circular symmetry of the magnetic field allows us to write  $\vec{B} \cdot d\vec{l} = B \cdot dl$ .

Assume that when strings are initially located at distance  $h$  above the center of the coil and let us set this point as the origin of our reference frame as shown in Fig.3. When string is plucked, it is located at an arbitrary position  $(x(t), y(t))$  at time  $t$ . Then the magnetic field at the center of the coil is now a function of time:

$$r(t) = \sqrt{x^2(t) + (y(t) + h)^2} \quad (2)$$

Assuming  $B(r(t))$  is almost constant across the surface of the coil, it is customary to define the effective area of the coil,  $A_{eff}$ , to simplify calculations. Then the total flux through the coil becomes

$$\phi(t) = \int_S \vec{B} \cdot d\vec{S} = B(r(t)) = \frac{\mu_0 I A_{eff}}{2\pi \sqrt{x^2(t) + (y(t) + h)^2}} \quad (3)$$

Once a string is plucked it will vibrate causing an ever-changing flux. We know from Faraday's law [1] that changing flux will induce a voltage between the terminals of the coil. Also taking into account Lenz's law, total voltage induced in  $N_C$  windings is

$$\begin{aligned} \mathcal{E}_i(t) &= -N_C \frac{d\phi(t)}{dt} \\ &= -\frac{\mu_0 I A_{eff} N_C}{2\pi} \left[ \frac{x(t)\dot{x}(t) + (h + y(t))\dot{y}(t)}{(x^2(t) + (y(t) + h)^2)^{\frac{3}{2}}} \right] \quad (4) \end{aligned}$$

where  $\dot{x}(t), \dot{y}(t)$  represents the time derivatives of strings coordinates (2) or speed of the cable in  $x$  and  $y$  directions respectively.

Since both ends of the strings are fixed, it will form a standing wave of the form

$$\begin{aligned} f(z, t) &= A \sin(k_n z) \cos(\omega_n t) \\ k_n &= \frac{n\pi}{L} \quad , \quad \omega_n = \frac{n\pi}{L} \sqrt{\frac{T}{\mu_{mass}}} \quad (5) \end{aligned}$$

where  $A$  is the amplitude of the wave,  $T$  is tension on the string,  $L$  is length of the string,  $\mu_{mass}$  is the mass density of the wire and  $n$  is an integer determining the order of the harmonic [2]. Since we are only interested about the point coil is located along the string, we set  $z = z_0$  constant and lump the  $\sin(k_n z_0)$  with  $A$  as  $\tilde{A}$  denoting the amplitude of the oscillations at  $z_0$ . Moreover, we assume that horizontal displacement of the string is negligible,  $x(t) \approx 0$ , to simplify calculations and set  $y(t) = f(z_0, t)$ . Then (4) transforms into

$$\mathcal{E}_i = -\frac{\mu_0 I A_{eff} N_C}{2\pi} \frac{\dot{y}(t)}{(y(t) + h)^2} = \mathcal{K} \frac{\omega_n \sin(\omega_n t)}{(h + \tilde{A} \cos(\omega_n t))^2} \quad (6)$$

where  $\mathcal{K} = \frac{\mu_0 I A_{eff} N_C \tilde{A}}{2\pi}$  is a lumped constant determined by the system parameters. Induced voltage is not a pure sinusoidal as one may demand. However, for sufficiently large  $h$ ,  $\tilde{A} \cos(\omega_n t)$  will be negligible and induced voltage will nearly be sinusoidal with the same frequency as mechanical vibrations. This means waveform of the sound waves (if there was a medium for them to propagate) will be preserved which is exactly what we require from a transducer. As can be seen from Fig. 4, waveform resembles to sinusoidal with increasing values of  $h$ . Even if the distance  $h$  is low, waveform can be smoothed out by isolating the fundamental frequency via low pass filters.

Strings must be continuously supplied with current to generate a magnetic flux in the pick-up coil. As a result, a power dissipation of  $P_{loss} = (10A)^2 \times 1\Omega = 100W$  occurs through the  $1\Omega$  resistor. However, this energy loss can be completely eliminated in the absence of resistive elements, which is feasible in the vacuum of space. The temperature in empty space is approximately  $2.7K$ , which is close to absolute zero. At such low temperatures, many

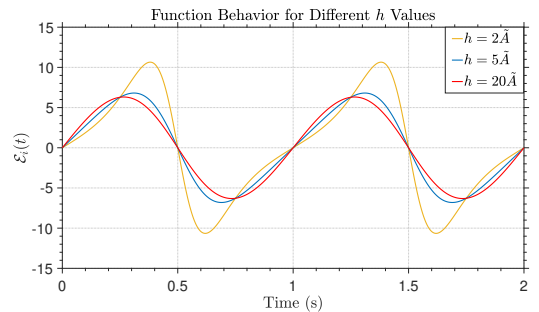


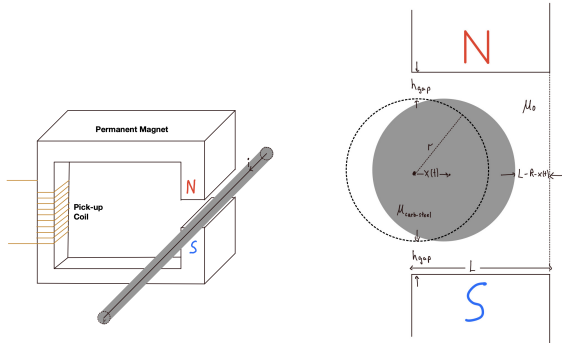
Fig. 4: Waveform of the induced voltage for different  $h$  values with  $\omega_n = 2\pi$

materials, including carbon steel and copper, enter a state of superconductivity, where their resistance drops to zero.

Leveraging this property, we introduce an additional path to the carbon steel wire, incorporating a switch at the junction of the wire with battery (see Fig. 2). Initially, the switch is set to position 1, ensuring that the battery supplies a total current of 10A, which is evenly distributed across the five carbon steel strings, resulting in 2A per string. After a brief period, the astronaut shifts the switch to position 2, connecting the auxiliary superconducting wire. Once this occurs, the current continues to circulate autonomously within the loop formed by the five plucked strings and the auxiliary wire. Since superconductors exhibit zero resistance, the current perpetually circulates without requiring an external source. Consequently, no additional current flows from the battery, thereby eliminating resistive power losses in the system.

### B. Design Option II: Permanent Magnet

Design option discovered in this subsection is similar to the previous one in the sense that mechanical motion of the string is detected via the changes in the magnetic flux in the pick-up coil. However, second design option involves the use of a permanent magnet. A permanent magnet can create a steady magnetic field without the need for a continuous electric current. Changes in the position of the ferromagnetic string reflects upon pick-up coil as induced voltage. This design was heavily inspired from the pick-up mechanism that converts string vibrations in electroguitars to electrical signals [3].



(a) Schematic of design option 2 using permanent magnets. (b) Close-up view of the air-gap.

As seen from Fig. 5a, a permanent magnet is placed around each string providing a homogeneous, linear magnetic field in the region carbon steel wire can be practically found in. Also copper wire is wrapped around the magnetic core to detect flux changes. We know that magnetomotive force (mmf) created by the magnet will induce a flux

$$\mathcal{F} = \phi \mathcal{R} \quad , \quad \mathcal{R} = \frac{l_c}{\mu A} \quad (7)$$

where  $\mathcal{F}$  is mmf,  $\phi$  is the total flux and  $\mathcal{R}$  is the reluctance [4]. Total reluctance of the system will be the sum of reluctances of magnetic core, air-gap and carbon-steel wire.

Since permeability of the magnetic core is very high ( $\mu \rightarrow \infty$ ), its reluctance is negligible. We assume that magnetic field is only effective only on the perpendicular cross-sectional area of the magnets and sharply drops to zero outside this area, neglecting any fringing effects for simplicity. Furthermore, we exclude the nonlinearities arising from the circular geometry of the wire and define an effective area to compensate for nonlinear effects. We further assume that mean path length,  $l_c$ , of the carbon-steel wire is always equal to its diameter  $2r$  and mean path length of the air-gap is also  $2r$ .

Total reluctance will be equal to the reluctance of the string in parallel with the reluctance of the air-gap of length  $2r$  and a series reluctance of the air-gap of mean length  $2h_{gap}$  at the edges of gap as can be seen from the Fig. 5b.

$$\begin{aligned} \mathcal{R}_{tot} &= \mathcal{R}_{string} || \mathcal{R}_{gap-2r} + \mathcal{R}_{gap-h} \\ &\cong \left( \frac{1}{\frac{2r}{\mu A_{eff} \frac{(r+x(t))}{L}}} + \frac{1}{\frac{2r}{\mu_0 A_{eff} \frac{(L-r-x(t))}{L}}} \right)^{-1} + \frac{2h_{gap}}{\mu_0 A_{eff}} \end{aligned} \quad (8)$$

where  $x(t)$  is the horizontal instantaneous displacement of the string at time  $t$ ,  $A_{eff}$  is the effective area of the whole cross-section of the magnet,  $\mu$  is magnetic permeability of the carbon-steel wire and  $\mu_0$  is magnetic permeability of air. Note that we have defined the area of the wire as in proportion to how far cable stretches out horizontally into the vertical cross-section of the magnet and the area of the air-gap with  $2r$  path length as the remaining from it. Carrying out the calculations, we get

$$\begin{aligned} \mathcal{R}_{tot} &= \left( \frac{\mu_0 A_{eff}}{2r^2} (\mu_r r + \mu_r x(t) + L - r - x(t)) \right)^{-1} \\ &\quad + \frac{2h_{gap}}{\mu_0 A_{eff}} \\ &\cong \frac{2r^2}{\mu_0 A_{eff}} \frac{1}{(x(t) + r)(\mu_r - 1) - L} \end{aligned} \quad (9)$$

where  $\mu_r = \frac{\mu}{\mu_0}$  is the relative permeability of the string. Note that it's safe to neglect the last term since  $h_{gap} \ll r$  and it merely is a constant which will be eliminated when we take the derivative of the flux to find the induced voltage. As the horizontal position of the string changes, total reluctance changes and so does the flux going through the pick-up coil. We find the induced voltage in the coil as

$$\begin{aligned} \phi &= \mathcal{F} \frac{\mu_0 A_{eff}}{2r^2} [(x(t) + r)(\mu_r - 1) - L] \\ \mathcal{E}_i &= -\frac{d\phi}{dt} = \mathcal{F} \frac{\mu_0 A_{eff}(\mu_r - 1)}{2r^2} \frac{dx(t)}{dt} \end{aligned} \quad (10)$$

Now replacing  $x(t)$  with the wave equation (5). We get the induced voltage similar to previous section II-A as

$$\mathcal{E}_i = -\frac{d\phi}{dt} = \mathcal{F} \frac{\mu_0 A_{eff} \tilde{A}(\mu_r - 1)}{2r^2} \omega_n \sin(\omega_n t) \quad (11)$$

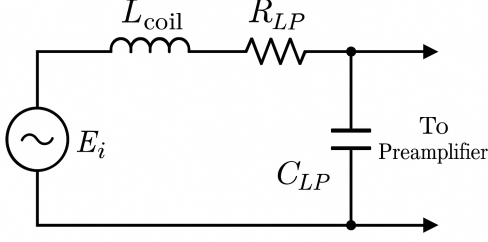


Fig. 6: Reduced circuit before preamplifier circuit.

Again, we can successfully realize the a sinusoidal vibration in the string as an electrical signal. Also note that amplitude of the induced voltage is proportional to the frequency of oscillations in the string. This means that our instrument will tend to amplify high frequency sounds while attenuating bass sounds which is undesirable. Nevertheless, it is an inevitable result of using an electronic element in the circuit and this effect is present even in real life electro-guitar pick-ups [5].

### C. Post Induction Circuit

We have discussed two methods for transducing the mechanical vibrations in the string as electrical signals. We have to now take into account the resistance, capacitance and the inductance as well. Also we utilize a low pass filter at the next stage to eliminate high frequency noise and vibrational frequencies human ear can not perceive in order to ensure a low transmission bandwidth and thus a more efficient system.

Ignoring intermediate connection cables that are used between electrical component in the circuit; resistance, capacitance and inductance of the pick-up coil is calculated as

$$\begin{aligned} R_{coil} &= \rho_0 \frac{\ell}{A} = 20 \mu\Omega, \\ L_{coil} &= \mu_0 N^2 \pi r^2 / \ell_{coil} \approx 39.4 \mu H, \\ C_{coil} &\approx \pi \epsilon_0 \ell / \ln(d/r_{wire}) \approx 16.7 \text{ pF} \end{aligned}$$

where  $\rho_0 = 0.02 n\Omega \cdot m$  is used for resistivity of copper at 0 K and separation of wires is assumed to be equal to the diameter  $d = 2r_{wire}$ . For most practical scenarios, values of  $R_{coil}$  and  $C_{coil}$  is too small to be significant and thus, we neglect them. Taking into account the inductance of the coil and the RC low-pass filter at the next stage, electronic circuit now looks like as in 6 where  $E_i$  is the induced voltage in the pick-up coil. Transfer function of this circuit can be given by

$$\begin{aligned} \frac{E_o(s)}{E_i(s)} &= \frac{\frac{1}{jsC_{LP}}}{R_{coil} + jsL_{LP} + \frac{1}{jsC_{LP}}} \\ &= \frac{\frac{1}{L_{coil}C_{LP}}}{s^2 + 2s\frac{R_{LP}}{L_{coil}} + \frac{1}{L_{coil}C_{LP}}} \end{aligned} \quad (12)$$

which is a second-order low-pass filter with cutoff frequency  $\omega_c = \frac{1}{\sqrt{L_{coil}C_{LP}}}$  [6]. We should choose the capacitance of the filter so that cutoff frequency of the filter is the upper limit of audible frequency range which is approximately 20kHz. A quick calculation yields  $C_{LP} = 0.634 \mu H$  for the capacitance of low pass filter. Note that this low pass characteristic also compensates for the effect that magnitude of the high frequency sounds are amplified while lower frequency sounds are attenuated as mentioned in II-B.

### III. DISCUSSION AND RESULTS

In conclusion, this project has explored the design of a string instrument that enables sound production in outer space using electromagnetic principles. Out of the two design options considered, first one uses magnetization of carbon steel strings to produce an electric flux at the pick-up coil while second uses permanent magnets the create this mmf. Either way, core principle we have utilized was the Faraday's law which states that changing flux caused by the mechanical motion of the strings will induce a voltage which will then be picked up by the copper coils.

We want to highlight that although analysis considered a purely sinusoidal vibration, operation of the device is not limited to solely single frequency signals. It has been proven that instruments output is a shifted and scaled version of the input sinusoidal with an arbitrary frequency, effectively deeming the device an LTI system. Therefore, any vibrational waveform on the string can be converted to electrical signal with a minor potential distortion since each signal can be decomposed to it's frequency component.

Moreover, both of the design options introduced are very energy efficient in the sense that transducer don't constantly need an external energy source to operate. In II-A, temperature of the space is utilized with the addition of a switch which allows the initially generated current to perpetually loop through superconducting carbon-steel wires without any additional energy. In II-B, we don't even need a battery to start with since permanent magnets are used to create the magnetic field. Also incorporation of the low-pass filter introduced in II-C significantly reduces the bandwidth of the signal by attenuating the frequencies beyond the audible range, resulting in less energy consumption by the transmitter and receiver units.

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